Fiscal Year:	FY 2017	**Task Last Updated:**	FY 03/28/2017
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Project Title:	Rapid Signaling Changes Induced by Gravity in Cells of the Fern Ceratopteris richardii		
Division Name:	Space Biology		
Program/Discipline:	SPACE BIOLOGY		
Program/Discipline--Element/Subdiscipline:	SPACE BIOLOGY--Cellular and molecular biology		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Cell & Molecular Biology (2) Plant Biology		
Space Biology Cross-Element Discipline:	(1) Developmental Biology		
Space Biology Special Category:	None		
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Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2012 Space Biology NNH12ZTT001N
Start Date:	08/01/2013	**End Date:**	12/31/2016
No. of Post Docs:	1	**No. of PhD Degrees:**	1
No. of PhD Candidates:	2	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	3	**Monitoring Center:**	NASA KSC
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Flight Program:			
Flight Assignment:	NOTE: End date changed to 12/31/2016 per NSSC information (Ed., 5/26/16)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Clark, Greg Ph.D. (University of Texas)		
Grant/Contract No.:	NNX13AM54G		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	Single-celled spores of the fern Ceratopteris richardii respond within seconds to both the direction and magnitude of gravity stimuli by altering the direction and magnitude of a trans-cell Ca2+ current that is required for the gravity response. This proposal aims to clarify the molecular mechanisms by which gravity alters the trans-cell current, and identify key signaling steps that link the current to cellular polarity changes directed by gravity. This focus on gravity signaling addresses an overarching question in the Space Biology Science Plan. Ca2+-uptake channels and Ca2+-ATPase efflux pumps drive the Ca2+ current across spore cells. Transcriptomic and proteomic analyses of genes and proteins highly expressed in spores at the peak of the Ca2+ current have identified likely candidates for the pumps and channels. Immunological approaches and yeast-based activity assays will be used to determine whether these transporters are localized on the plasma membrane and are regulated post-translationally. Recent evidence indicates that extracellular eATP [eATP], a newly discovered signaling agent, helps control the gravity response in Ceratopteris, just as it helps regulate gravitropism in Arabidopsis roots. Because eATP controls the activity of Ca2+ channels in plants, we will test the regulatory link between eATP and the activity of Ca2+ channels that is crucial for the gravity response in Ceratopteris. Ca2+-binding proteins such as calmodulins, annexins, and Ca2+-dependent protein kinases are highly expressed in Ceratopteris at the peak of the Ca2+ current. They are the most likely signal transducers that help mediate the downstream cell polarization events guided by gravity, and their regulatory role in the gravity response will be tested. Expected results will generate hypotheses that predict molecular responses likely to occur in Ceratopteris cells in response to the micro-g environment, and the proposal will describe International Space Station (ISS) experiments that could test these predictions.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The germinating spore of Ceratopteris is an especially valuable model system for studying the cellular bases of gravity-induced graviresponses on Earth, because both the sensing of gravity and the response occur in the same cell. The cellular level mechanisms of sensory responses are very similar in plant and animal cells, thus the bottom-to-top Ca2+ current that develops early during spore germination as one of the first polarized events induced by gravity in the cell may be an early event also in animal cells known to respond to gravity, such as muscle and bone cells. Our investigation into candidate molecules likely to be associated with both the efflux and influx phases of this current will help clarify how this current is controlled. A number of studies have documented the importance of Ca2+ currents for polarized growth in both animal and plant cells; however, there are few prior studies that have identified and analyzed the molecular components of these currents. Moreover, while it is generally accepted that Ca2+ is a second messenger that participates in a myriad of plant and animal sensory responses, there is still little known about the molecular identity of the Ca2+ channels that mediate many of these responses, especially responses to gravitational stimuli. The proposed tests of Ca2+ pump and channel function and regulation, including possibly channel regulation by eATP, will not only provide novel insights into how the gravity-directed Ca2+ current originates, but also further advance an understanding of channel structures. The results will set the stage for future assays of post-translational changes that occur during space flight in the Ca2+-ATPase pumps and/or mechanosensitive channels in fern and possibly also in some animal cells when the transporter activities go from near zero in micro-gravity to quite high as the g-force rises to 2-g or higher levels achieved during parabolic flight or on a centrifuge on the International Space Station (ISS).
	The specific aim of the novel research proposed was to discover the molecular bases of rapid gravity responses in single spore cells of the fern Ceratopteris richardii. The approach to achieve this goal was to identify from recently obtained transcriptomic data molecular components that would be likely to drive the calcium current that is rapidly modulated by gravity. This would then allow an assessment of post-transcriptional signaling changes in these components that are both induced by gravity and needed for subsequent, gravity-directed polarity changes in cell division and growth. Prior literature suggests many of the post-transcriptional changes induced by gravity in single Ceratopteris cells are similarly induced in multi-cellular organs of flowering plants, so the findings of this project are likely to have relevance to gravitational studies in the more complex tissues of higher plants. Expected findings from the proposed studies would predict rapid molecular-level changes that will be induced in the micro-g environment, and testing the validity of these predictions in Space Flight experiments will be both feasible and likely to provide valuable new insights on the systems life uses to sense and react to gravity. One of the goals of Aim 1 of the original proposal was to molecularly identify cation transporters that could play a major role in driving the trans-cell calcium current that is directed by gravity in fern spores cells. In accord with this goal, we obtained the full-length sequence of a putative MscS (mechanosensitive channel, small current) gene, and obtained evidence in support of the conclusion that it actually has Msc activity by showing that it could partially rescue a bacterial knockout of Msc. We also obtained partial sequences of two other Msc candidates in Ceratopteris that will serve as the basis to design primers that can “walk” through the Ceratopteris transcriptome and genome to obtain their full sequences, and the primary sequences of the proteins they encode. This will enable the production of antibodies that can be used to identify where each of the Msc channels is localized. If any of these candidates prove to be mechanosensitive channels on the plasma membrane, it (or they) would be the one(s) most likely to undergo post-translational modifications by gravity stimulation. The goal of Aim 2 of the original proposal was to test the role of genes encoding Ca2+-binding proteins that are expressed at the peak of the trans-cell Ca2+ current by suppressing or overexpressing their transcription using transformation methods. We have successfully generated transgenic Ceratopteris spores expressing one of several different transgenes, and these spores will be valuable tools for genetically dissecting gravity responses in spores that respond to gravity while they are still single cells. The mutants include one expressing the calcium reporter Yellow Cameleon, which will enable assessing gravity-induced changes in [Ca2+]cyt; one constitutively expressing a spore annexin, which is a calcium-binding protein implicated in regulating calcium channels in plasma membranes, one suppressed in annexin expression; one knocked down in expression of the Ca2+ATPase pump that helps drive the gravity-induced trans-cell calcium current; and one knocked down in one of the mechanosensitive ion channels described above in the paragraph on Aim 1. Generating these mutants and genetically confirming their mutant status required over two years of ground-breaking research, and the transgenic products are the first (and still the only) fern mutants modified in spore-expressed genes. We made the most progress on Aim 3 of the original proposal, which was to test the role of eATP in promoting the critical Ca2+-uptake step of the spore gravity response. Key background information that provides the rationale for this goal is that when gravity directs the polarization of Ceratopteris fern spores this process begins with the uptake of calcium through channels at the bottom of the spore, and this step is necessary for the gravity response. Data from several different laboratories showing that extracellular ATP (eATP) regulates calcium channels in plants and animals

Task Progress:	led to the hypothesis that extracellular nucleotides could play a role in the gravity-directed polarization of Ceratopteris spores. In animal and plant cells ATP can be released from mechanosensitive channels. The idea that a differential distribution of a signaling molecule could lead to morphological polarity is not unique. In this regard, gradients of the growth hormone auxin are among the best documented to be inductive of cell polarity in plants. The differential distribution of eATP during gravity-directed polarization could be an early step in a series of asymmetrical molecular events that ultimately lead to the downstream cellular and morphological changes needed to generate a gravity-directed, downward growing rhizoid in Ceratopteris spores. Precedent data relevant to this hypothesis are those of Tang et al. (Wenqiang Tang, Shari R. Brady, Yu Sun, Gloria K. Muday, and Stanley J. Roux. Extracellular ATP Inhibits Root Gravitropism at Concentrations That Inhibit Polar Auxin Transport. Plant Physiol. 2003 131: 147-154.) in the journal Plant Physiology, who showed that a high [eATP] inhibits the gravity response in Arabidopsis roots. They interpreted their data as evidence that an eATP gradient could be one of the first steps involved in the gravity response in plant cells. These data led us to test whether gravity could induce a gradient of eATP between the top and bottom of germinating fern spores, and, if so, whether this gradient could help mediate the gravity effect of directing the polarization of the spores. In order to carry out this test, an ATP biosensor, developed by the McLamore laboratory at the University of Florida, was used to measure the [eATP] at the bottom and top of germinating spores during gravity-directed polarization. This biosensor could quantify eATP levels down to the low nM range, and it has a tip diameter of only 1-2 µm, much smaller than the 100-120 µm diameter of spore cells. Using this unique tool with a micromanipulator that could guide the probe to within a few microns of the spore surface, we made the exciting discovery that the [eATP] along the bottom of the spore averaged 7-fold higher than the concentration at the top. Treatments that disrupted the gradient, such as flooding the medium with ATP, or that antagonized eATP receptors, such as adding the ATP receptor blocker PPADS to the medium, or that hydrolyzed ATP in the medium, such as by the addition of phosphatase enzymes, all resulted in a statistically significant decrease in the gravity response. In order to investigate the source of ATP release, spores were treated with Brefeldin A (BFA), which would block cellular secretory activity, and gadolinium trichloride (GdCl3), which would block the transport activity of the mechanosensitive channels that release ATP into the extracellular medium. These treatments resulted in a significant decrease in gravity-directed polarization. The same eATP biosensor was also used to measure ATP release after treatment with both BFA and GdCl3. Both of these treatments caused a significant decrease in [ATP] measured around spores. These results supported the hypothesis that ATP could be released from mechanosensitive channels and secretory vesicles during gravity-directed polarization. The novel findings described above show a clear role for eATP in gravity-directed polarization of Ceratopteris spores by documenting that gravity can induce a bottom-to-top gradient of extracellular ATP around the spores, and that this gradient promotes gravity-directed polarization of the spores. They provide support for the hypothesis that mechanosensitive channels, which have been implicated in other reports as critical components of gravity responses, can help generate eATP gradients that can enhance a cellular gravity response. Overall, they reveal that quantifying and evaluating gradients of eATP can lead to a more thorough understanding of how plants use ATP release to direct growth and development. Production of the novel self-referencing biosensor used to document the gravity-induced eATP gradient between the top and bottom of vertically-oriented spore cells would not have been possible without support from this NASA grant. Recognizing that this biosensor would be a valuable research tool for investigating other eATP-regulated phenomena not only in ferns but also in higher plants, we used it to assay [eATP] levels around growth zones of primary roots of Arabidopsis, because previous research had indicated cells release ATP as they grow. We also used the probe to evaluate whether changing the expression of two apyrase (NTPDase) enzymes (AtAPY1 and AtAPY2), which had been implicated in hydrolyzing and thus limiting the [eATP], actually altered the [eATP]. Our results, which we presented in posters at two international meetings, showed that the [eATP] is highest in root zones that have the highest growth rate (i.e., elongation zones) and that the constitutive expression of AtAPY1 or AtAPY2 lowered this concentration, whereas suppression of these apyrases raised the [eATP]. These results provided novel and important verification of previously postulated hypotheses about the relationship of eATP and apyrase enzymes to growth, and they are being assembled into a manuscript that we expect to submit this summer.
Bibliography Type:	Description: (Last Updated: 10/16/2023)
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